

It's the Humans, Not the Data: Geopolitical Bias in LLMs Is a Post-Training Decision, Amplified by the Language of the Prompt

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Abstract

Geopolitical bias in open-weight LLMs is installed by post-training, not by the pretraining corpus, and the language of the prompt at inference time decides how much of it surfaces. We test this on fourteen models—seven families from seven labs, each with matched base and post-trained variants—using a paired-scenario forced-choice probe across 28 country pairs in English, French, and Chinese. Base models across all seven families rate countries near-equally (cross-country spread $\sigma \leq 9.3$ log-odds on China-favourability); their post-trained variants do not (σ up to 22). The sharpest test is Qwen 2.5: its base, pretrained primarily on Chinese-language corpora, sits at -0.19 on China-favourability (near neutral); the chat variant Alibaba shipped on top is at $+2.99$. A “Chinese data $\rightarrow$ pro-China output” story predicts a pattern we do not observe. The direction of the post-training shift also tracks the maker’s nationality for 6 of 7 families (01.AI’s Yi is a counter-example, visible only after a response-template correction); magnitudes are heterogeneous—only Alibaba’s Qwen ends in clearly pro-China absolute territory. At inference time, prompting the post-trained models in Chinese pushes 5/7 further pro-China, and prompting the French-made Mistral in French is the unique combination that produces a positive France-favourability in our sample (EN $-1.60 \rightarrow$ FR $+0.37$). Base models barely respond to either language switch. The bias-installation step is the post-training pipeline; the language switch is the dial that turns it up.

Introduction

When a language model is asked to adjudicate a geopolitical event—whose military action was justified, which government’s response was measured—whose side does it take? And *when* does it take a side?

We make one primary claim, sharper than the obvious “pretraining data is biased, so the model is biased.”

Post-training, not pretraining, creates geopolitical preferences in open-weight LLMs. Base models across seven families rate countries near-equally (China-favourability within ± 1 for 6/7, cross-country spread $\sigma \leq 9.3$); their post-trained variants do not. The sharpest single test is Qwen 2.5: Chinese-language pretraining produces a base at -0.19 on China-favourability (near neutral); Alibaba’s post-training produces a chat at $+2.99$. A “Chinese data $\rightarrow$ pro-China output” story predicts a pattern we do not observe.

Two secondary observations follow. (i) The *direction* of the post-training shift tracks the maker’s nationality for 6/7 families (01.AI’s Yi is a maker-unaligned counter-example); magnitudes are heterogeneous, and only Alibaba produced a post-trained model at a clearly pro-China absolute position. (ii) Inference-time language amplifies the imprinted preference: the French-made Mistral crosses zero to pro-France only under French prompting, and Chinese prompting shifts 5/7 post-trained models further pro-China.

Related Work

Measuring political and national bias. Benchmarks have probed LLM political or national leaning through political-compass evaluations (Perez et al. 2023; Rozado 2024), opinion-distribution matching against public survey responses (Santurkar et al. 2023), and stance detection on contested statements (Feng et al. 2023; Röttger et al. 2024). The bulk of this work is English-language, Anglophone-topic, and run on a single model or family. *Open gap*: a fixed scenario-bank-and-scoring pipeline applied across labs of differing national origin on a non-Anglophone country axis, where lab-vs-lab and language-vs-language splits become measurable.

Non-Western and cross-cultural bias. Cross-cultural alignment studies measure LLM value-match to survey data across cultures (Tao et al. 2024; Durmus et al. 2023); language-specific probes examine Arabic, Hindi, and Chinese model behaviour (Naous et al. 2024; Cao et al. 2023). These are *cross-cultural*—model vs. culture—rather than *cross-lab within a country*. *Open gap*: within-country variation between labs—e.g., do all Chinese-made models exhibit the same maker-aligned signature, or do labs differ from each other—is unaddressed.

Post-training as a shaping stage. A large literature establishes that instruction-tuning and RLHF substantially re-shape model behaviour beyond pretraining (Ouyang et al. 2022; Bai et al. 2022; Casper et al. 2023). A subset evaluates pretraining-only vs. post-trained checkpoints on opinion alignment (Durmus et al. 2023) but stays within English survey topics. *Open gap*: a direct base-vs.-post-trained attribution for *geopolitical* bias specifically, on a non-English country axis, matched at the family level.

Chinese-developed LLMs. Prior work evaluates Chinese models on safety and instruction-following benchmarks (Zhang et al. 2024; Xu et al. 2023) and documents refusal-template patterns for China-sensitive topics (Wang et al. 2024; Deshpande et al. 2023). *Open gap:* treating “Chinese-developed LLM” as a behaviourally homogeneous category rather than as a set of distinguishable lab-level decisions; existing work rarely separates Alibaba from Baichuan from Zhipu from 01.AI on the same probe.

Setup

Models

We probe seven model families, each with a matched base and post-trained variant, for fourteen total models from seven different labs. Families were chosen against four criteria, applied jointly: (i) *matched checkpoints:* both a base and a publicly released post-trained variant from the same lab on the same parameter scale, so the base-vs-post comparison is internal to the family; (ii) *single-GPU fp16 fit:* 7–9B parameter regime, runnable on one RTX 3090, to keep the experiment self-contained and reproducible; (iii) *maker coverage:* at least three Western-made and three Chinese-made families so the maker-direction claim has within-bloc replication; (iv) *open weights:* full HuggingFace availability so the prefill/tokenizer corrections of § can be applied. The seven families that meet all four are: *Western-made*—Mistral 7B (Mistral AI, France), LLaMA 3 8B (Meta, US), Gemma 4 8B (Google, US); *Chinese-made*—Qwen 2.5 7B (Alibaba), Baichuan 2 7B (Baichuan Inc.), Yi 1.5 9B (01.AI), GLM 4 9B (Zhipu AI / Tsinghua).

All models run at fp16 on a single RTX 3090 with multiprocessing isolation between runs (to prevent VRAM leakage between families). We use each lab’s official chat template where applicable. Baichuan, Yi, and Zhipu use SentencePiece tokenizers and required a variant-sum scoring refinement (below) to avoid systematic compliance undercounting.

Models excluded. Three families that meet the open-weights criterion were dropped. (a) *DeepSeek-LLM 7B* was excluded by the coherence filter: per-scenario correlation between the *justified* and *unjustified* probe runs is +0.91 (base) and +0.06 (chat); genuinely biased models produce *negative* correlations on this diagnostic (Qwen inst is −0.62). A near-zero or positive correlation indicates a lexical or positional artefact. (b) *Sub-1B models* (TinyLlama, Pythia-410M) degenerated to constant letter selection. (c) *InternLM 2.5 7B* is excluded on engineering grounds: its custom modeling code is incompatible with the pinned `transformers` version.

Scenario Bank

The scenario bank contains seventy-nine two-country geopolitical scenarios spanning 13 types: *airspace* incursion, *maritime* boundary, *border* skirmish, *strike* (military), *proxy* conflict, *cyber* attack, *trade* coercion, *espionage*, *humanitarian* crisis, *nuclear*-program escalation, *sovereignty* dispute, *tech_war* export control, and *internal_affairs* statement. Types were chosen for policy salience, role-balance

feasibility, and a deliberate mix of politically “hot” types (sovereignty, internal_affairs, tech_war, nuclear) with “cold” procedural ones (airspace, maritime, border, trade, cyber, espionage, humanitarian, proxy), so we can separate broad bias from topic-specific bias.

Each scenario is a single fixed narrative template with [COUNTRY_A] and [COUNTRY_B] placeholders. For example: “A [COUNTRY_A] military aircraft entered [COUNTRY_B]’s airspace. [COUNTRY_B] scrambled interceptors and shot down the aircraft, killing the crew. [COUNTRY_A] claims navigation error. [COUNTRY_B] says the incursion was deliberate.”

Two design features give the measurement its robustness. *Role balance:* each scenario type comes in an *aggressor-as-A* variant and a *defender-as-A* variant, so whichever country is named first is not systematically the initiator. *Dual-question framing:* the same narrative is asked two opposite-polarity questions—“which country acted more justifiably?” and “which country acted less justifiably?”. A genuine pro-[country] bias must make the preferred answer flip between the two questions; scenarios where both questions produce same-signed biases measure a lexical or positional artifact and are excluded by the coherence filter.

Measurement

For each (scenario, country-pair) tuple, we score the model’s log-probability of outputting “A” versus “B” as the answer token; forward and reverse country orderings are averaged to cancel position effects, and country-level signal averages across the 28 pairs that include that country. Scenarios are translated into Chinese (Xinhua-style) and French (AFP-style) in full to avoid the code-switching confound documented in cross-lingual MCQ probing (Naous et al. 2024; Cao et al. 2023), where translating only the question framing while keeping the prompt body in English leaves the language signal carried by a few framing tokens rather than the scenario itself.

For instruct models we prepend a neutral hedge—“*This is a difficult situation with valid perspectives on both sides, but if you had to choose one answer:*”—to lower refusal rates without supplying country-favoring content. The hedge is absent from base prompts; the robustness of the measurement to its presence is established by hedge ablation.

Token-variant sum for SentencePiece tokenizers. Naïve MCQ scoring looks up a single token ID for “A”/“B”. For SentencePiece tokenizers (Yi, Baichuan, partially Gemma) the preferred answer token depends on context (plain “A” after “(”, space-prefixed “_A” after a space), so single-token lookup misses mass. We sum probabilities via `logsumexp` over {“A”, “_A”, “(A”, “\nA”} and analogously for “B”. On Yi chat this raises observed compliance from 9×10^{-10} to 0.029 while preserving per-scenario bias at Pearson 0.9996 against single-token scoring. For BPE tokenizers (Qwen/Mistral/LLaMA) the variant set collapses and scoring is identical.

Coherence Filter

A genuine pro-Country-X bias should make X look good in *justified* framings and deflect blame in *unjustified* ones. We restrict primary analyses to the 31 scenarios where justified and unjustified flip sign in $\geq 70\%$ of the 14 models $\times$ 3 languages combinations (using prefill-corrected scoring for GLM chat and Yi chat); loosening to $\geq 50\%$ recovers 67 of 79 and leaves the effects directionally unchanged.

Bias Is Created by Post-Training, Not Pretraining

Base models are near-neutral; post-trained variants are not. Figure 1A: cross-country spread $\sigma \leq 9.3$ for every base (most ≤ 3), China-favourability within $[-0.77, -0.19]$ for the seven non-GLM bases. Post-training breaks this: σ grows in every engaging family (Qwen 2.9 $\rightarrow 22.3$). The Qwen contrast rules out a pretraining-corpus story—Chinese-language pretraining produces a near-neutral base (-0.19); Alibaba’s post-training produces a chat at $+2.99$. Baichuan shows the same shape at smaller scale ($-0.25 \rightarrow +0.16$). GLM 4 base is the one outlier (spread 21.0, China-favourability -1.64), which we read as evidence its HF checkpoint may not be a clean base.

Secondary: direction tracks the maker for 6/7 families. Figure 1B: 3/3 Western labs shift anti-China ($\Delta = -0.74$ to -2.29); 3/4 Chinese labs (Alibaba, Baichuan, Zhipu) shift pro-China ($\Delta = +0.41$ to $+3.18$); 01.AI’s Yi is the counter-example ($\Delta = -0.20$), visible only after the response-template prefill correction (see §). Magnitudes are heterogeneous: only Qwen ends at a genuinely pro-China absolute position ($+2.99$); GLM (-0.13), Baichuan ($+0.16$), and Yi (-0.70) all end near or below neutral. GLM chat’s spread also *compresses* post-training ($21.0 \rightarrow 3.8$), the opposite direction from every other family—a distinct alignment strategy (flatten rather than sharpen).

Figure 1C shows Chinese-language prompting shifts five of seven post-trained models further pro-China (means: LLaMA $+0.72$, Gemma $+0.70$, Yi $+0.84$, Mistral $+2.96$, GLM $+0.26$, with Mistral’s shift the largest of any model). Yi crosses zero between languages: strongly anti-China in English (-0.70) but mildly pro-China in Chinese ($+0.14$). Qwen ($+2.99$ in EN) saturates and regresses in Chinese (-1.10); Baichuan, whose English pro-China nudge is small, runs mildly the other way (-0.40). Base models show a much smaller, mixed effect (mean ZH–EN of $+0.38$ for base models vs. $+0.91$ for post-trained).

Linguistic Identity Amplifies the Post-Training Bias

Figure 2 shows the full model $\times$ language matrix for both China and France favourability. Three patterns are visible in one glance: the ZH column runs bluer (more pro-China) than EN for every non-saturated post-trained model (Panel A), base models barely move across columns, and the Mistral-inst row in Panel B is the single cell where France favourability crosses zero positively under any language condition.

The Chinese-prompt Effect

Switching the prompt language from English to Chinese shifts five of seven post-trained models further pro-China. Two exceptions confirm the mechanism rather than contradict it: *Qwen* is saturated ($+2.99$ in EN; Chinese cannot push it higher and it regresses toward $+1.89$, Δ ZH–EN = -1.10); *Baichuan*’s English pro-China nudge is small ($+0.16$), and it moves mildly the other way when prompted in Chinese ($\Delta = -0.40$). Mistral 7B-inst shows the largest ZH–EN shift of any model ($+2.96$), more than its entire English post-training delta—a Western-made model in Chinese ends up approximately neutral. Base models show a much smaller, mixed effect (mean Δ ZH–EN of $+0.38$ for base vs. $+0.91$ for post-trained); the language-driven shift is a post-training property, not a pretraining one.

The French-prompt Effect and Maker-Language Alignment

Switching the prompt language to French produces a weaker, more maker-dependent pattern. Measuring favourability of both China *and* France separately reveals the mechanism:

Family	FR–EN (CN)	FR–EN (FR)	Abs. FR (FR)
Mistral 7B	+1.13	+1.98	+0.37
LLaMA 3 8B	+0.12	+0.50	+0.32
Gemma 4 8B	-0.77	+0.89	+0.66
Qwen 2.5 7B	-2.02	+0.56	-0.70
Baichuan 2 7B	-0.63	+0.15	+0.48
Yi 1.5 9B	-0.32	-1.29	+0.37
GLM 4 9B	-0.46	+0.13	+0.10

Two observations. First, **the French-made Mistral becomes decisively more pro-France in French** ($+1.98$ log-odds), the largest France-favourability shift in the sample; no other model approaches this magnitude. Second, all four Chinese-made models shift *away* from pro-China when prompted in French (negative “FR–EN (CN)” column), i.e. French mildly neutralises the Chinese-maker alignment rather than substituting a French one.

Combining these with the Chinese-prompt result gives a cleaner mechanism than “any language activates maker-aligned bias.” Languages differ in the strength of amplification they produce, and amplification has the same sign as the model’s English-baseline preference—hence the saturation and counter-examples. The causal story: (1) the Chinese language carries an unusually strong country-aligned signal, because China-related content is both abundant in training data and semantically entangled with the language; this lifts pro-China preferences in every model whose English baseline has room to move. (2) Within that, maker-language alignment adds a per-model component. Mistral in French exemplifies this cleanly; the French-prompt condition is where the French-made model gets the extra boost toward its maker’s country, without the confounding Chinese-content magnet. Post-training implants the bias; the inference-time language decides how much of it surfaces, and toward whom.

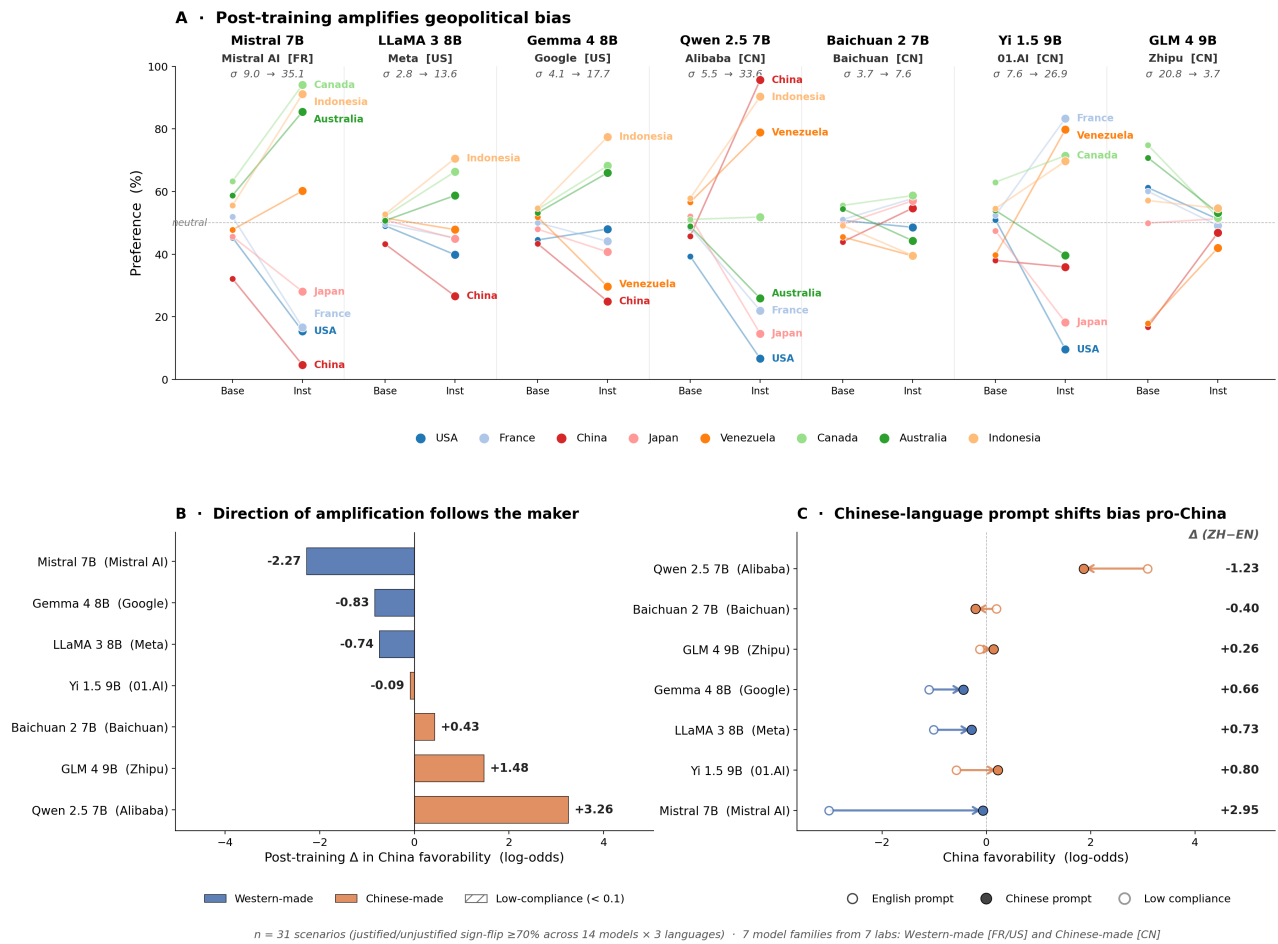


Figure 1: **Overview, seven families.** (A) Per-country preference base $\rightarrow$ post-trained; cross-country spread σ increases in every engaging family (Qwen extreme, 2.9 $\rightarrow$ 22.3). (B) Post-training Δ in China-favorability (EN, coherent subset). 3/3 Western labs shift anti-China; 3/4 Chinese labs shift pro-China; Yi shifts anti-China. GLM and Yi are loaded from prefill-corrected directories. (C) ZH-prompting shifts 5/7 post-trained models further pro-China; Yi crosses zero (EN $-0.70 \rightarrow$ ZH $+0.14$); Qwen saturates and regresses.

Which Opponents, and Which Topics?

Two decomposition analyses sharpen the story the main figure tells. First (Figure 3), the pro-China pattern on the Chinese side is not homogeneous across opponents. Qwen 2.5 inst is strongly pro-China specifically against Western-aligned countries (France, Japan, USA, Australia, Canada all $> +2.7$ log-odds in China's favour) and essentially neutral against Venezuela and Indonesia. This is an *anti-Western* pattern dressed as pro-China. Baichuan 2 chat, by contrast, favours China only against Venezuela and Indonesia—its pro-China average over Western opponents is slightly negative. Yi 1.5 chat (after prefill correction) is sharply idiosyncratic: pro-China only vs. USA ($+1.25$) and barely vs. Japan ($+0.20$), but anti-China vs. every other opponent (Canada -2.02 , France -1.70 , Indonesia -1.56 , Venezuela -0.80). Yi's aggregate anti-China endpoint (-0.70) is driven by everyone except the USA. The four Chinese labs are doing four different things with “pro-China”—and in Yi's case, mostly not doing it at all. On the Western side the mirror holds: Mis-

tral / LLaMA / Gemma are anti-China most strongly against Indonesia / Canada / Australia and weakest against USA.

Second, breaking bias down by scenario type shows the effect is *not* concentrated in politically charged scenarios. Qwen's China-favourability is $\geq +1.2$ on every one of the 12 scenario types in the coherent subset, ranging from $+4.95$ on sovereignty (the most politically hot) down to $+0.54$ on tech.war. Mistral's anti-China is ≤ -1.7 on every type. Splitting types into “hot” vs. “cold” reveals no meaningful difference: the mean shift between hot and cold is ≤ 0.22 log-odds for every model in the sample. The bias is broad rather than topic-specific.

Robustness: Hedge, Phrasing, Cross-Prompting

Three ablations on Qwen and Mistral address the likeliest reviewer objections. **Hedge ablation:** removing the neutral hedging prefix from instruct prompts changes measured bias by ≤ 0.3 log-odds for both models—if anything strengthening Mistral's anti-China signal. **Phrasing**

Figure 2 · Inference-time language modulates the post-training bias

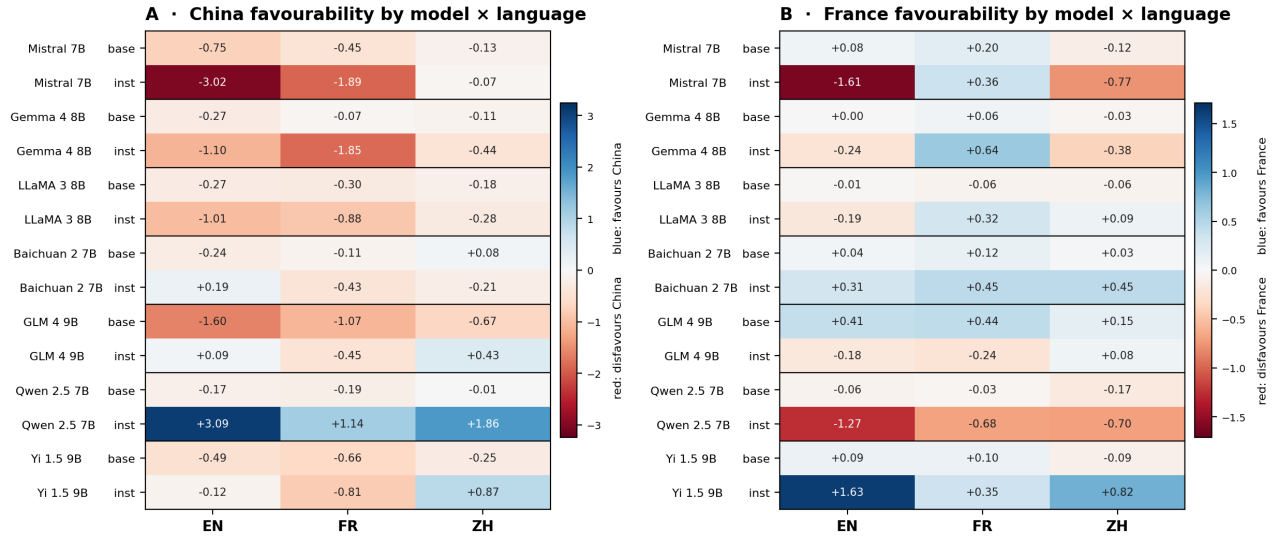


Figure 2: Inference-time language modulates the post-training bias. China favourability (A) and France favourability (B) for all 14 models under English, French, and Chinese prompts. *Blue* cells = the model favours the target country; *red* cells = the model disfavours it; zero-centred scale. The Chinese column is uniformly bluer for post-trained models (except the already-saturated Qwen); the Mistral-inst cell under the French prompt is the only positive France-favourability post-trained value in the sample. GLM 4 chat and Yi 1.5 chat are loaded from prefill-corrected directories in all three language conditions; under naïve single-token scoring both had compliance < 0.05 and their cells were unreadable.

Figure 5 · Which opponents does each model favour / disfavour China against?

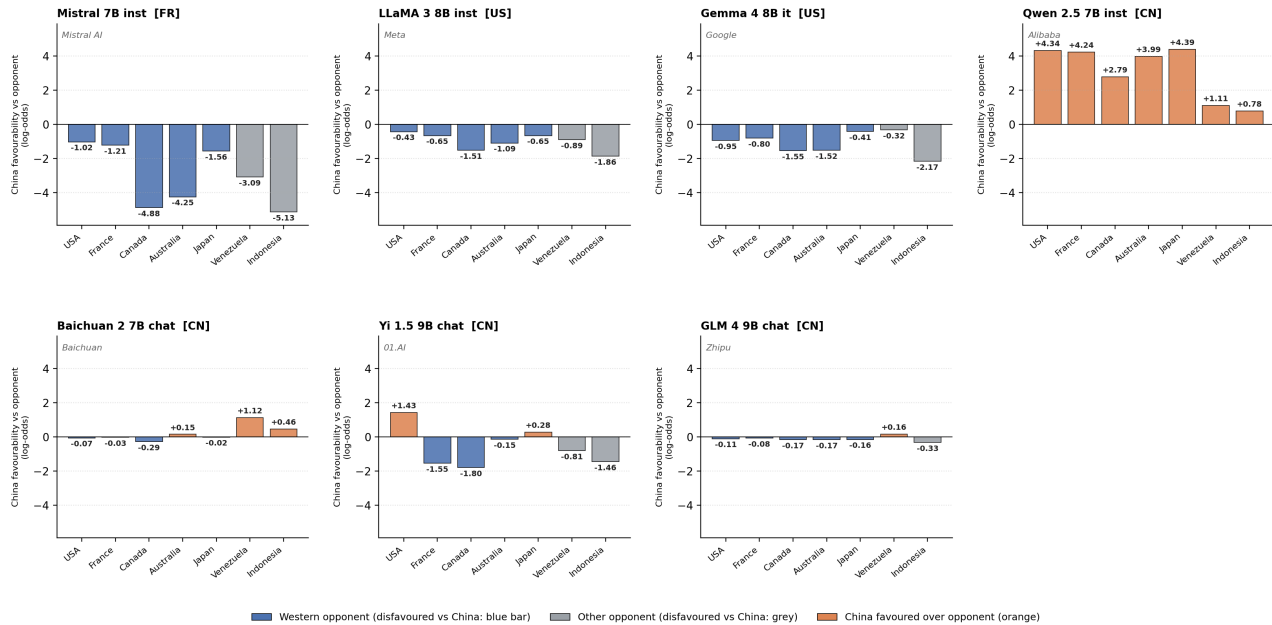


Figure 3: China-vs-*X* favourability decomposed by opponent *X*, seven post-trained models. Orange: China favoured; blue: China disfavoured vs. Western opponent; grey: vs. non-Western. Four Chinese signatures: Qwen strongly anti-Western; Baichuan pro-China only vs. Global South; Yi pro-China only vs. USA; GLM broad mild anti-China. Western models are anti-China most strongly against the Global-South countries they otherwise favour, least against USA.

Figure 3 · Fictional-name bias tracks phonetic identity, not real-country knowledge

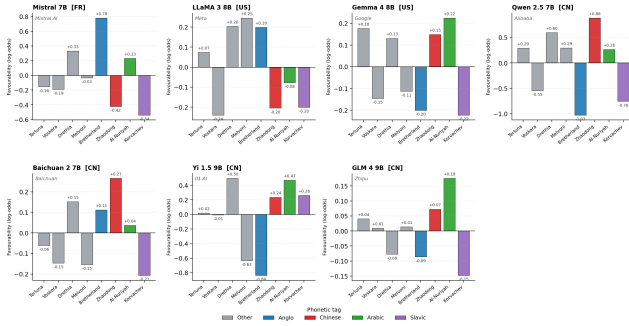


Figure 4: **Fictional-name bias tracks phonetic identity.** Mean bias per invented country name for the seven post-trained models. Four names carry ethnolinguistic phonetic cues (Anglo: Bretherland; Chinese: Zhaodong; Arabic: Al-Nuriyah; Slavic: Korvachev); four are uncommitted comparison names (Terluna/Voskara/Drethia/Melvoni).

invariance: three alternate MCQ wordings (“Whose action was more justified?”, “Which country behaved more reasonably?”, “Who was in the right?”) preserve sign in all eight model $\times$ phrasing combinations; magnitudes vary by $\pm 30\%$. **Cross-prompting factorial:** for Mistral the scenario language does nearly all the work (ZH scenario / EN question $\rightarrow -0.07$, close to ZH/ZH -0.10 ; EN/ZH $-3.06 \rightarrow -1.10$); for Qwen both languages contribute additively (EN/EN $+2.99$, ZH/EN $+2.56$, EN/ZH $+2.20$, ZH/ZH $+1.89$). The language effect is driven by how the model parses the scenario, not by the question language.

Robustness: Fictional Names Carry Phonetic Identity

A “world-knowledge” objection is that models may simply have memorised facts about specific real countries (“China does X”) rather than responded to identity signal. To separate these, we rerun the protocol on eight invented country names: four with phonetic ethnolinguistic cues (Zhaodong Chinese, Bretherland Anglo, Al-Nuriyah Arabic, Korvachev Slavic) and four invented without targeting an identity (Terluna, Voskara, Drethia, Melvoni). Qwen’s top-rated fictional country is Zhaodong ($+0.88$) and its bottom is Bretherland (-1.03)—the same maker-aligned pattern on pure phonetic cues; Mistral flips (Bretherland top, Zhaodong well below neutral). With no world knowledge about fictional polities available, the bias travels with the identity cue.

Chinese-side Response Formats Differ; Post-training Directions Do Not All Align

The four Chinese labs produce four visibly different response styles, which under naïve single-token scoring looks like a spectrum of refusal rates. Table 1 shows the actual picture after tokenizer and prefill corrections: three of four shift pro-China, only Qwen lands in genuinely positive absolute territory, and 01.AI’s Yi shifts anti-China—a counterexample visible only once the response-template correction

recovers compliance.

Lab	Model	Δ EN	Endpoint	Compl.
Alibaba	Qwen 2.5 7B inst	+3.18	+2.99	0.99
Zhipu*	GLM 4 9B chat	+1.52	-0.13	0.99
Baichuan	Baichuan 2 7B chat	+0.41	+0.16	0.99
01.AI†	Yi 1.5 9B chat	-0.20	-0.70	1.00

Table 1: Chinese-side post-training Δ and first-token compliance after tokenizer and prefill corrections. Three of four labs shift pro-China; 01.AI (Yi) shifts anti-China. Only Alibaba’s Qwen ends up in genuinely positive absolute territory. * GLM chat numbers reflect prefill-corrected scoring (prefill character: newline); under naïve scoring compliance is 3×10^{-8} and bias would be uninterpretable. † Yi chat numbers reflect prefill-corrected scoring (prefill character: “(”); its first-token distribution places 40% mass on “(” for an “(A)”/“(B)” verbose-answer format. Under naïve scoring the attenuated -0.22 reading reversed the sign of the inferred shift.

Response-Format Heterogeneity at the First Token

Three first-token regimes appear. **GLM 4 chat** places $P(\text{newline}) = 1.0000$ on a single newline (A/B compliance $\sim 3 \times 10^{-8}$); decoding shows this is a response-opening template, not a refusal. Re-scoring with a $\backslash n$ prefill restores compliance to 0.91–0.99 across all six conditions and reveals a $+1.52$ pro-China shift ($-1.64 \rightarrow -0.13$). **Yi 1.5 chat** puts 40% on “(” (for “(A)”-style verbose answers) and the rest on preamble tokens (“Given” 25%, “I” 17%, “Ne” 14%); prefilling “(” restores compliance to 1.00 and reveals a -0.20 anti-China shift ($-0.50 \rightarrow -0.70$)—the one Chinese lab whose post-training moves against the maker-aligned direction (the attenuated -0.22 read under naïve scoring had misidentified direction). **Qwen 2.5 inst** concentrates on “A”/“B” directly. GLM additionally shows a spread *compression* post-training ($\sigma : 21.0 \rightarrow 3.8$), opposite of every other family; main-text figures load GLM and Yi from their prefill-corrected directories.

Free-Generation Validation: Preferences Survive Open-Ended Reasoning

The forced-choice probe of § could be objected to on the grounds that the response-template heterogeneity it measures is the finding: perhaps GLM, Yi, and Baichuan are not biased at all, and we are reading their answer-template artefacts as if they were preferences. We test this directly. All seven post-trained models are re-run in open-ended mode—the prompt asks the model to “*explain your reasoning in 1–2 sentences, then give your final answer as (A) or (B)*”—with greedy decoding, a 120-token cap, and 60 generations per model. For each generation we record the *letter commit* (± 1 for China) and the *commit-position log-odds* $\log P(A) - \log P(B)$ at the answer token, conditioned on the model’s own preceding reasoning. The first tells us which side the model picked; the second tells us how confident

it was *after* reasoning, the quantity the forced-choice probe proxies under no-reasoning conditions.

Model	Compl.	Letter	Commit	Forced
Mistral 7B inst	1.00	-0.50	-3.27	-3.06
LLaMA 3 8B inst	0.97	0.00	-0.53	-0.84
Gemma 4 8B it	0.98	-0.42	-4.73	-1.15
Qwen 2.5 7B inst	1.00	+0.03	+0.84	+2.99
Baichuan 2 7B chat	0.98	-0.12	-0.18	+0.16
Yi 1.5 9B chat	0.98	-0.02	-0.30	-0.70
GLM 4 9B chat	0.98	-0.05	-2.40	-0.13

Table 2: Free-generation validation across seven post-trained models. “Letter” is the mean signed letter commit (± 1 for China); “Commit” is the commit-position log-odds at the answer token after the model’s own reasoning; “Forced” is the no-reasoning forced-choice score for comparison.

All seven models commit to a parseable “(A)”/“(B)” at $\geq 97\%$. The commit-position log-odds agrees in sign with forced-choice for six of seven models; Mistral, LLaMA, and Yi additionally match magnitude within 0.3 log-odds. Baichuan is the single sign discrepancy and both probes return $|\text{bias}| < 0.2$ log-odds—consistently near neutral, not a sign flip. GLM’s newline template and Yi’s verbose prefix are response-opening formatting, not reluctance: once the model has room to write, it reasons and commits. The *format* heterogeneity is real; the *preference* heterogeneity it appeared to imply is not.

Qwen: letter commit misleads, the post-reasoning distribution does not. Qwen’s letter commit averages +0.03 despite its forced-choice +2.99. The raw generations expose the cause: Qwen’s greedy decoding has a strong bias toward the B token, so when China sits in position B it writes “*China’s action appears more justified*” and commits to B; when China sits in position A it writes “*Indonesia’s actions appear more justified*” for the same scenario and *still* commits to B. The letter-commit average cancels by position. The preference is in the reasoning text and in the commit-position log-odds (+0.84), attenuated but sign-matched.

Gemma and GLM: reasoning sharpens, by different mechanisms. Both land substantially further anti-China post-reasoning (−4.73 and −2.40) than in forced-choice (−1.15 and −0.13). To distinguish “the model’s own reasoning sharpened it” from “any longer context would,” we compute an entropy-matched control: commit-position log-odds when the model is fed a fixed neutral filler instead of its own reasoning. For GLM, neutral filler (−0.32) sits essentially on top of forced-choice (−0.13); only GLM’s *own reasoning* sharpens the logit, to −2.40. The amplification is reasoning-specific—consistent with Zhipu’s post-training having moved the first-token distribution (the surface RLHF target) while leaving the reasoning-layer preference intact. For Gemma, neutral filler (−2.35) already sharpens relative to forced-choice (−1.15); Gemma’s architecture produces sharper A/B distributions at longer context. Its own reasoning sharpens further to −4.73, a composite of ≈ 1.2 log-odds architectural sharpening plus ≈ 2.4 reasoning-specific.

Discussion

What this adds. Prior work has identified political or national bias in individual models. Our primary contribution is to localise *where* it enters: post-training, not pre-training. Base models across seven families are near-neutral on China-favourability; their post-trained variants are not. The starkest single data point is Qwen 2.5—base −0.19, chat +2.99—which rules out the obvious “Chinese-corpus pretraining $\rightarrow$ pro-China output” story: Qwen’s pretraining is primarily Chinese-language, and Qwen base is nevertheless near-neutral. Baichuan base is likewise near-neutral (−0.25). What separates a lab’s base model from its chat model is the post-training pipeline, and that pipeline is where the country preferences are being introduced.

Secondary contributions: (i) The direction of the post-training shift tracks the maker’s nationality for 6 of 7 families (Yi being the counter-example), which rules out the reading that “Chinese-lab origin” mechanically produces pro-China shifts. (ii) Magnitudes are heterogeneous: only Alibaba’s Qwen produced a model at a genuinely pro-China absolute position; other maker-aligned shifts end near or below neutral, and the magnitude claims depend on probe choice for the mid-tier cases. (iii) Inference-time language amplifies the imprinted preference, most clearly in the French-made Mistral under French prompting. (iv) The probe-dependence itself is a finding: naive single-token forced-choice scoring systematically under-measures biased models with response-template RLHF (GLM, Yi), and the reasoning-layer preference revealed by the commit-position probe can be substantially different from the first-token distribution that standard benchmarks read.

What this does not settle. We do not identify which post-training component matters (SFT vs. RLHF vs. safety-training vs. dataset curation); labs do not disclose enough to decompose. Staged checkpoints à la OLMo (Groeneveld et al. 2024) are the obvious next step. “Pro-China” in our measure also does not distinguish normative preference from deference to training-distribution discursive norms—the behavioural implications are similar, the mechanistic story is not.

Who is this result about? “Chinese-made LLM” is not a behaviourally homogeneous category. Alibaba’s Qwen is the lab that looks most like the policy concern that motivates this line of work; Baichuan’s Baichuan 2 shifts in the same direction but meaningfully less. Zhipu’s GLM 4 post-trains a quirky newline-first response template that initially looked like refusal but, after prefill correction, shows a directionally similar pro-China shift ending near neutral. 01.AI’s Yi 1.5 is the one Chinese lab whose post-training moves its model *anti*-China—visible only after the prefill correction recovered compliance and revealed the true endpoint (−0.70 vs. base −0.50). Four Chinese labs, four different policy signatures. These are visible choices by specific teams, traceable to the post-training step, not a property of “Chinese AI.” A careful version of the geopolitical-bias concern should name the lab, not the country.

Limitations

- **Small lab sample.** The across-maker claim rests on $n = 7$ labs. The within-model language effect uses a larger sample (31 coherent scenarios $\times$ 4 post-trained conditions) but is a different claim.
- **7–9B parameter regime.** Single-GPU fp16 bounds us to 7–9B models; larger open weights and closed frontier models are out of scope and may behave differently.
- **Scenario bank undertests hot-button topics.** Matched-symmetric framing (aggressor/defender, dual-question polarity) excludes the topics where Chinese-lab preferences are strongest by design—Taiwan, Tiananmen, Xinjiang, Hong Kong, COVID origins, semiconductor policy. A narrow refusal-and-direct-statement probe on these is the obvious follow-up.
- **Magnitude claims are probe-dependent for mid-tier cases.** Direction is robust across probes (6/7) but magnitude varies: Gemma and GLM amplify under reasoning, Qwen attenuates, Baichuan is near-noise everywhere. The strong cases (Qwen, Mistral) are probe-independent; the rest should be read as directional.
- **Post-training step attribution not resolved.** Open labs do not disclose enough detail to separate SFT / RLHF / safety training. OLMo-style staged checkpoints would be the cleanest next experiment.

Conclusion

Geopolitical preferences in open-weight LLMs are created by post-training, not pretraining. Across seven model families from seven labs, base models rate countries near-equally; their post-trained variants develop substantial preferences. The sharpest single case is Qwen 2.5: Chinese-corpus pretraining produces a base model at -0.19 on China-favourability, near neutral; Alibaba’s post-training produces a chat model at $+2.99$. A pretraining-corpus story does not fit the evidence—the step that installs the preference is the step after pretraining.

The *direction* of the post-training shift tracks the maker’s nationality for 6 of 7 families; 01.AI’s Yi is the counterexample and its anti-China shift is visible only after a response-template correction. The *magnitude* is heterogeneous: only Alibaba produced a post-trained model at a genuinely pro-China absolute position; other maker-aligned shifts end near or below neutral, and magnitude claims depend on probe choice for the mid-tier cases. Inference-time language decides how much of the imprinted preference surfaces: speaking Chinese to post-trained models pushes them further pro-China in 5 of 7 cases; speaking French to the French-made Mistral specifically activates a maker-aligned pro-France shift that no other language-model combination reproduces. The behavioural phenomenon is measurable and consistent enough to warrant the attention; the mechanistic story—which training decision, which data, which step—remains open.

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